

Azure Monitoring

Azure Monitor is Microsoft's unified observability service for collecting, analyzing, and acting on telemetry from cloud and hybrid environments. It enables you to understand the health, performance, and reliability of your Azure applications and infrastructure resources by bringing together metrics, logs, traces, and events into a single observability experience.

Monitoring Data

→

Monitoring Data

Azure Monitor works with different types of telemetry data:

Metrics

Metrics are numerical values that are collected at regular intervals and describe some aspect of an Azure resource. They are automatically collected for most Azure resources at no cost.

Logs

A resource consists of event and log data collected from operating systems or applications.

Trace

It helps to understand the flow of a request and where performance issues arise in distributed systems by tracking the request's passage through various components.

The diagram illustrates the Azure Monitor architecture. On the left, four boxes represent data sources: 'Apps / Workloads', 'Infrastructure', 'Azure Platform', and 'Custom Sources'. An arrow points from these sources to a central box labeled 'Azure Monitor'. Inside this central box, there are four stacked sections: 'Metrics' (with a bar chart icon), 'Logs' (with a document icon), 'Traces' (with a document icon), and 'Insights' (with a lightbulb icon). To the right of the central box is another box labeled 'Observability agent' with a gear icon. An arrow points from the 'Observability agent' box to the 'Insights' section of the central box. The 'Observability agent' box also contains the text 'Explore Investigate Automate'. The 'Insights' section of the central box is further divided into four sub-sections: 'Application' (with a lightbulb icon), 'Kubernetes' (with a cluster icon), 'VM' (with a server icon), and 'Network' (with a globe icon). Below these are three more sub-sections: 'Visualize' (with a bar chart icon), 'Analyze' (with a bar chart icon), and 'Respond' (with a document icon). The 'Visualize' section contains 'Workbooks', 'Dashboards', 'Grafana', and 'Power BI'. The 'Analyze' section contains 'Metric Explorer' and 'Log Analytics'. The 'Respond' section contains 'Alerts', 'Actions', and 'Issues'.

Insights

Collect and analyze a subset of available telemetry for a given service(s). Depending on the service, the experiences might also provide out-of-the-box alerting. They present the telemetry in a visual layout. The visualizations vary in size and scale.

Visualize

It is the visualization of monitoring data collected by Azure Monitor into charts, dashboards, and reports using tools such as Workbooks, Dashboards, Grafana, and Power BI.

Analyze

For ad-hoc analysis, use metrics explorer to analyze metrics using PromQL and Log Analytics to analyze log and trace data using KQL.

Respond

Generating an alert when the conditions specified in the collected data are met ensures that the necessary actions are taken.

Create diagnostic settings for each of the following data types to send them to a Log Analytics workspace, archive them to a storage account, or stream them to an event hub for ingestion by services outside of Azure

Data type	Description	Data collection method
Activity log	Provides insight into subscription-level events for Azure services, including service health records and configuration changes.	Collected automatically. View in the Azure portal or create a diagnostic setting to send it to other destinations. You can collect it in a Log Analytics workspace at no charge.
Platform metrics	Numerical values automatically collected at regular intervals for different aspects of a resource. Specific metrics vary for each type of resource.	Collected automatically and stored in Azure Monitor Metrics. View in metrics explorer or create a diagnostic setting to send it to other destinations.
Resource logs	Provides insight into operations that were performed within an Azure resource. The content of resource logs varies by the Azure service and resource type.	You must create a diagnostic setting to collect resource logs.

Data Platform

Data Platform

Azure Monitor has a centralized data platform to support collection of telemetry from a variety of sources and the different features for analysis and troubleshooting. The platform uses two workspace types, each optimized for different data:

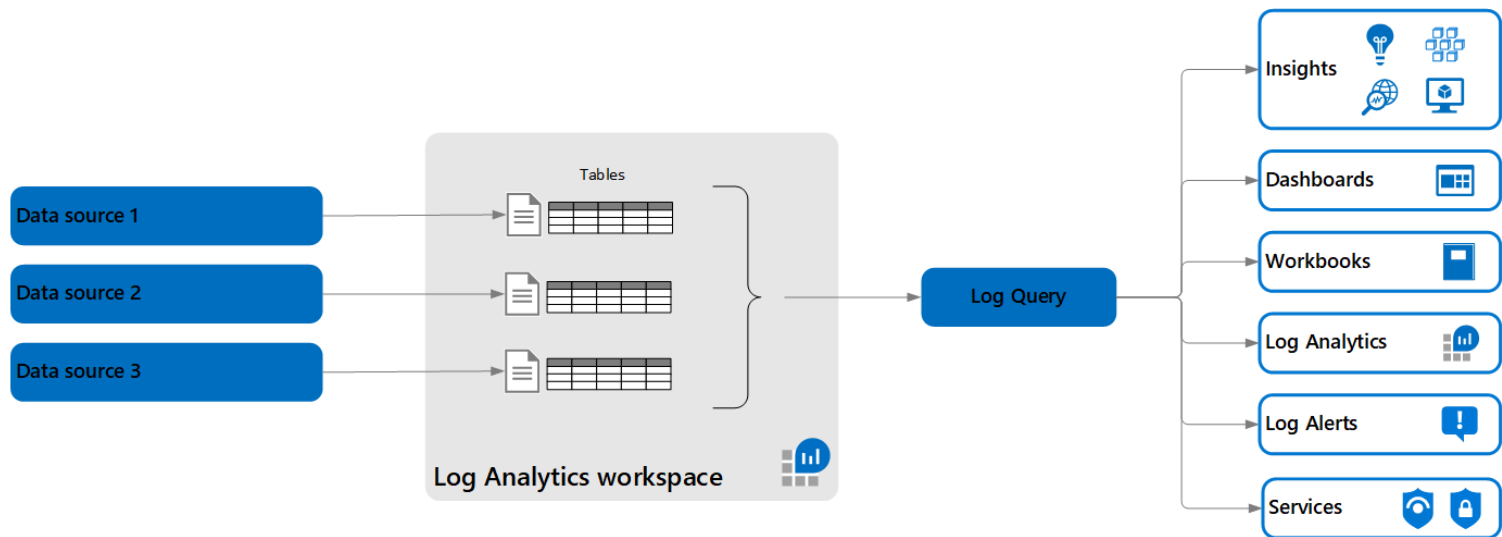
- Log Analytics workspaces**

A Log Analytics workspace is a data store into which you can collect any type of log data from all of your Azure and non-Azure resources and applications. Workspace configuration options let you manage all of your log data in one workspace to meet the operations, analysis, and auditing needs of different personas in your organization through.

Each Log Analytics workspace contains multiple tables in which Azure Monitor Logs stores data you collect.

Azure Monitor Logs automatically creates tables required to store monitoring data you collect from your Azure environment. You create custom table to store data you collect from non-Azure resources and applications, based on the data model of the log data you collect and how you want to store and use the data.

Table management settings let you control access to specific tables, and manage the data model, retention, and cost of data in each table.



collect log and trace data, which can be analyzed with Kusto Query Language (KQL).

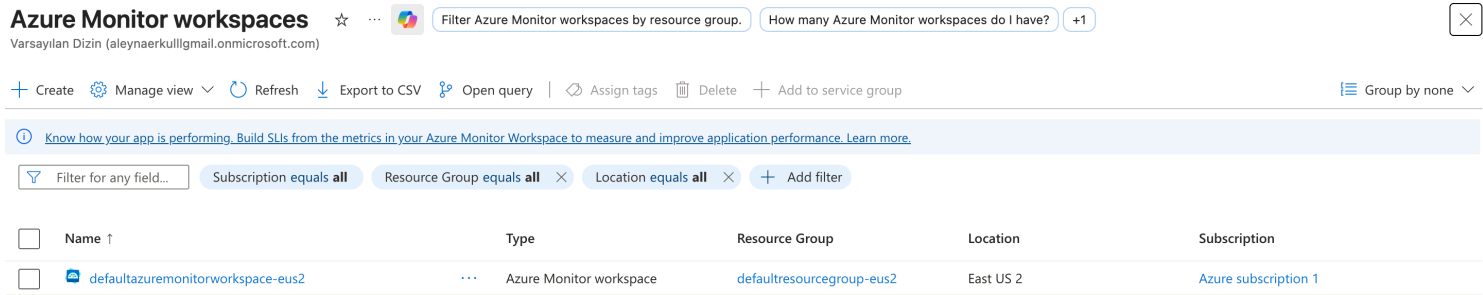
- Azure Monitor workspaces**

Azure Monitor workspaces will eventually contain all metric data collected by Azure Monitor. Currently, only Prometheus metrics are data hosted in an Azure Monitor workspace.(<https://learn.microsoft.com/en-us/azure/azure-monitor/metrics/azure-monitor-workspace-overview>)

These metrics can be queried using PromQL and visualized through Azure Monitor and related tools.

Example:

Guest OS → OpenTelemetry metrics → Azure Monitor Workspace → PromQL



collect Prometheus and OpenTelemetry metrics, which can be analyzed using Prometheus Query Language (PromQL).

Data Collection

Data Collection

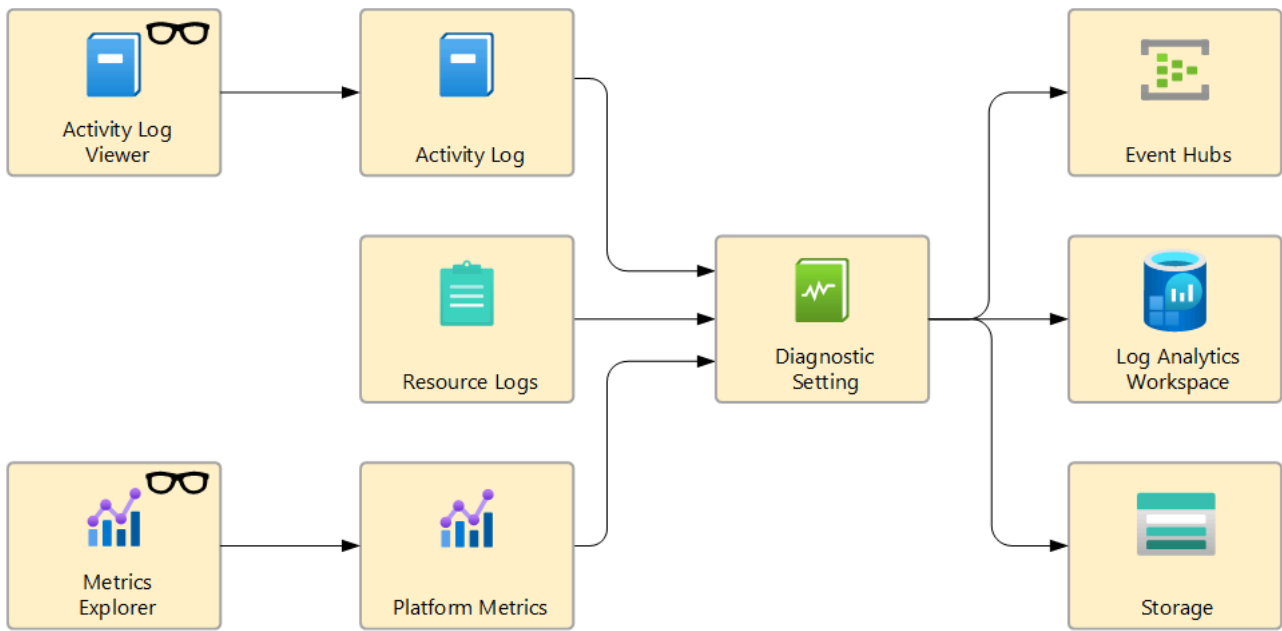
Azure Monitor collects monitoring data from various sources. Different data sources may use different data collection methods, and each may require different configurations. The collected data can be processed and subsequently sent to one or more destinations.

The main collection methods used in this study are:

- Diagnostic Settings**

You can use diagnostic settings in Azure Monitor to collect resource logs and to send platform metrics and the activity log to various destinations. Create a separate diagnostic setting for each resource that you

want to collect data from. Each setting defines the data from the resource to collect and the destinations to send that data to.



Home > Resource Manager | All resources > vnet_test | Diagnostic settings

Diagnostic setting

Save Discard Delete Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

Diagnostic setting name: vnet-ds

Logs

Category groups

- ☒ allLogs

Categories

- ☒ VM protection alerts

Metrics

- ☒ AllMetrics

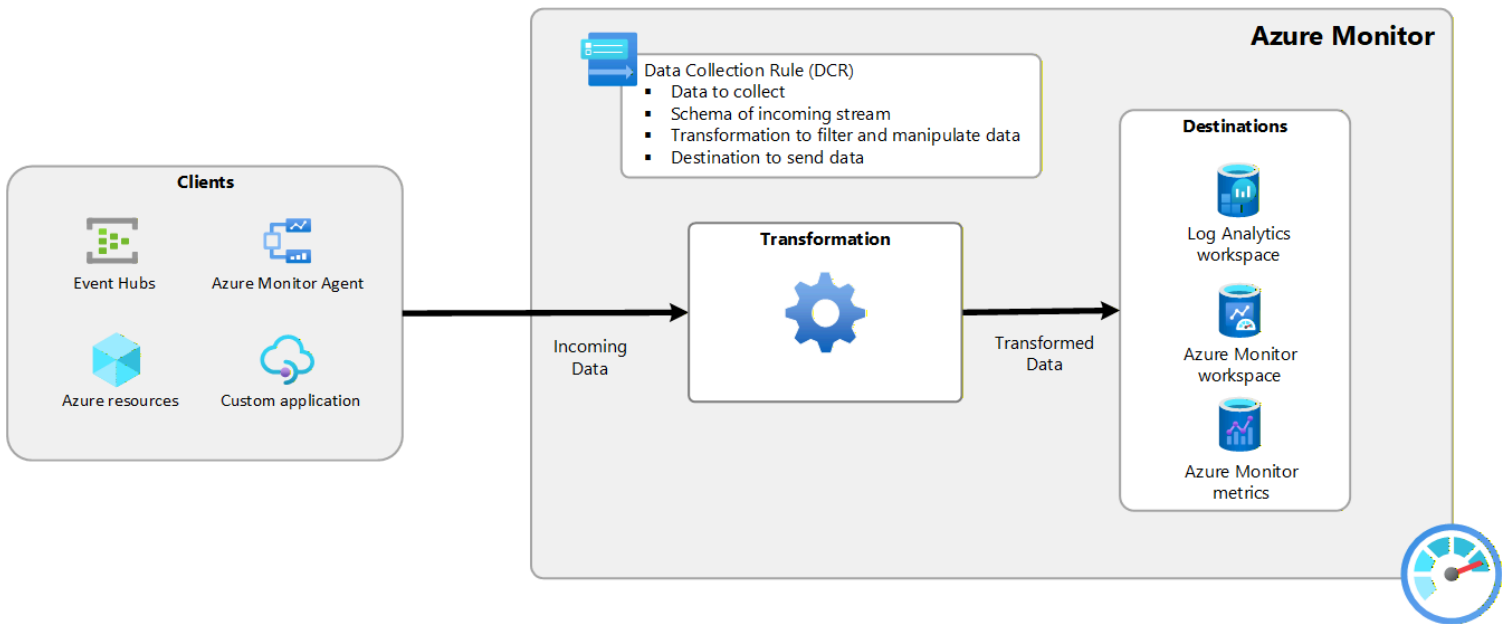
Destination details

- ☒ Send to Log Analytics workspace
- Subscription: Azure subscription 1
- Log Analytics workspace: DefaultWorkspace-2167be8b-b80b-4b66-8ca4-4d4dc2f111da-EUS (eastu...
- ☐ Archive to a storage account
- ☐ Stream to an event hub
- ☐ Send to partner solution

Data Collection Rules (DCRs)

The data collection process that DCRs support provides a common processing path for incoming data. Each data collection scenario is defined in a DCR. The DCR provides instructions for how Azure Monitor should process the data it receives. Depending on the scenario, DCRs specify all or some of the following items:

- Data to collect and send to Azure Monitor.
- Schema of the incoming data.
- Transformations to apply to the data before it's stored.
- Destination where the data should be sent.



After you create a DCR, use different methods to use it based on the data collection scenario. The following table lists common scenarios and the method used to collect data in each case. The following sections provide more details on each method.

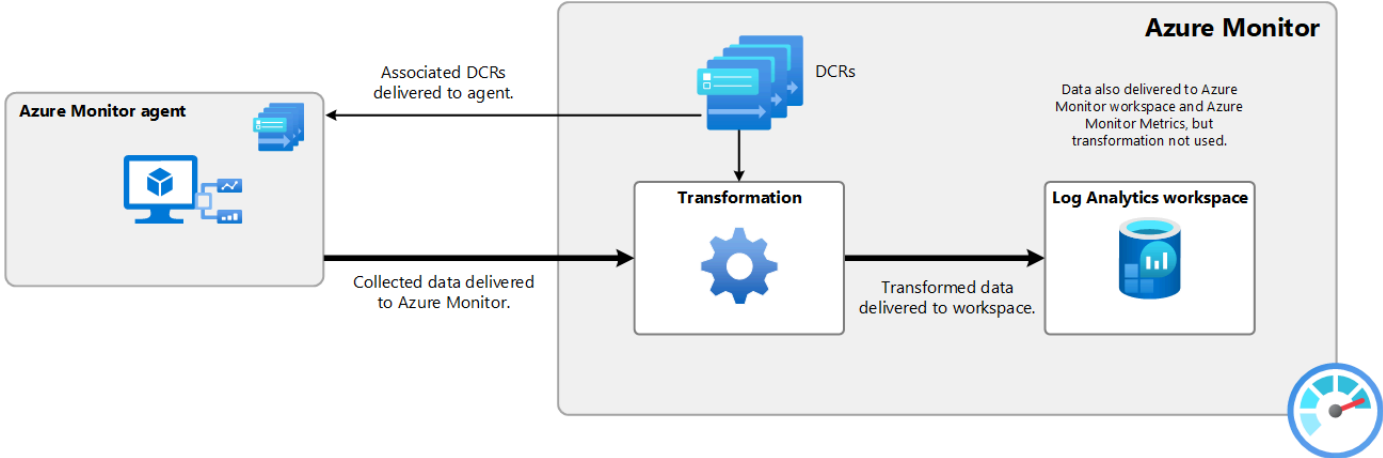
Scenario:

Azure Monitor Agent (AMA)

[The Azure Monitor Agent collects monitoring data from the guest operating system of Azure and hybrid virtual machines \(VMs\). It delivers the data to Azure Monitor for use by features, insights, and other services, such as Microsoft Sentinel and Microsoft Defender for Cloud.](#)

The Azure Monitor Agent collects data according to [Data Collection Rules \(DCRs\)](#) associated with the agent. DCRs define what data is collected, how it gets processed, and where it gets sent.

When you install the agent, it retrieves and applies any associated DCRs and periodically checks for updates. It enables centralized and consistent configuration of data collection across multiple agents and environments.



Event hubs

Azure Event Hubs is an event streaming service that ingests large volumes of data in real time and stores or forwards it so that other systems can access it.

When the Event Hub receives data, it forwards it to Azure Monitor. A Data Collection Rule Association (DCRA) is created between the Event Hub and the DCR. Azure Monitor then applies the transformation defined in the DCR to the data and sends it to the destination specified in the DCR.

Home > Event Hubs

Create Namespace

Event Hubs

BasicsAdvancedNetworkingTagsReview + create

Project Details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure subscription 1

Resource group *

rg-ocr-dev

Create new

Instance Details

Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).

Namespace name *

.servicebus.windows.net

Region *

East US 2

The region selected supports Availability zones. Your namespace will have Availability Zones enabled. [Learn more](#).

Pricing tier *

Basic

Browse the available plans and their features

Throughput Units *

1

Review + create

< Previous

Next: Advanced >

Choose your pricing tier

Browse the available plans and their features

Choose the best Tier for your intended purpose. [Learn more](#)

Tier	Pricing	Apache Kafka	Schema Registry	Max retention period	Storage retention	Extended retention	Geo-disribution	Maximum publish size
Billed per cluster on capacity units provisioned, incremental namespaces are not charged								
Dedicated ☆	Starting at ~\$6.849 USD per CU per Hour	Available	Available	90 days	10 TB/CU	Available	Available	20 MB
Billed per namespaces, based on number of units provisioned								
Premium ☆	Starting at ~\$1.027 USD per PU per Hour	Available	Available	90 days	1 TB/PU	Available	Available	1 MB
Standard	Starting at ~\$0.030 USD per TU per Hour	Available	Available	7 days	84 GB	Not available	Available	1 MB
Basic	Starting at ~\$0.015 USD per TU per Hour	Not available	Not available	1 day	84 GB	Not available	Not available	256 KB

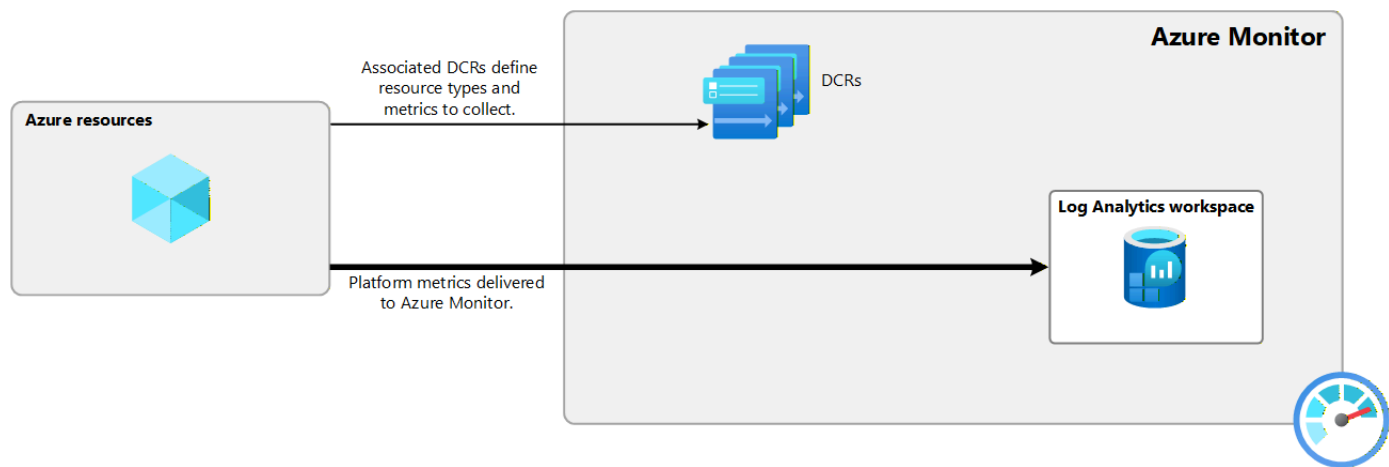
☆ Recommended

For complete list of Azure Event Hubs quotas and limits [click here](#)

Platform metrics

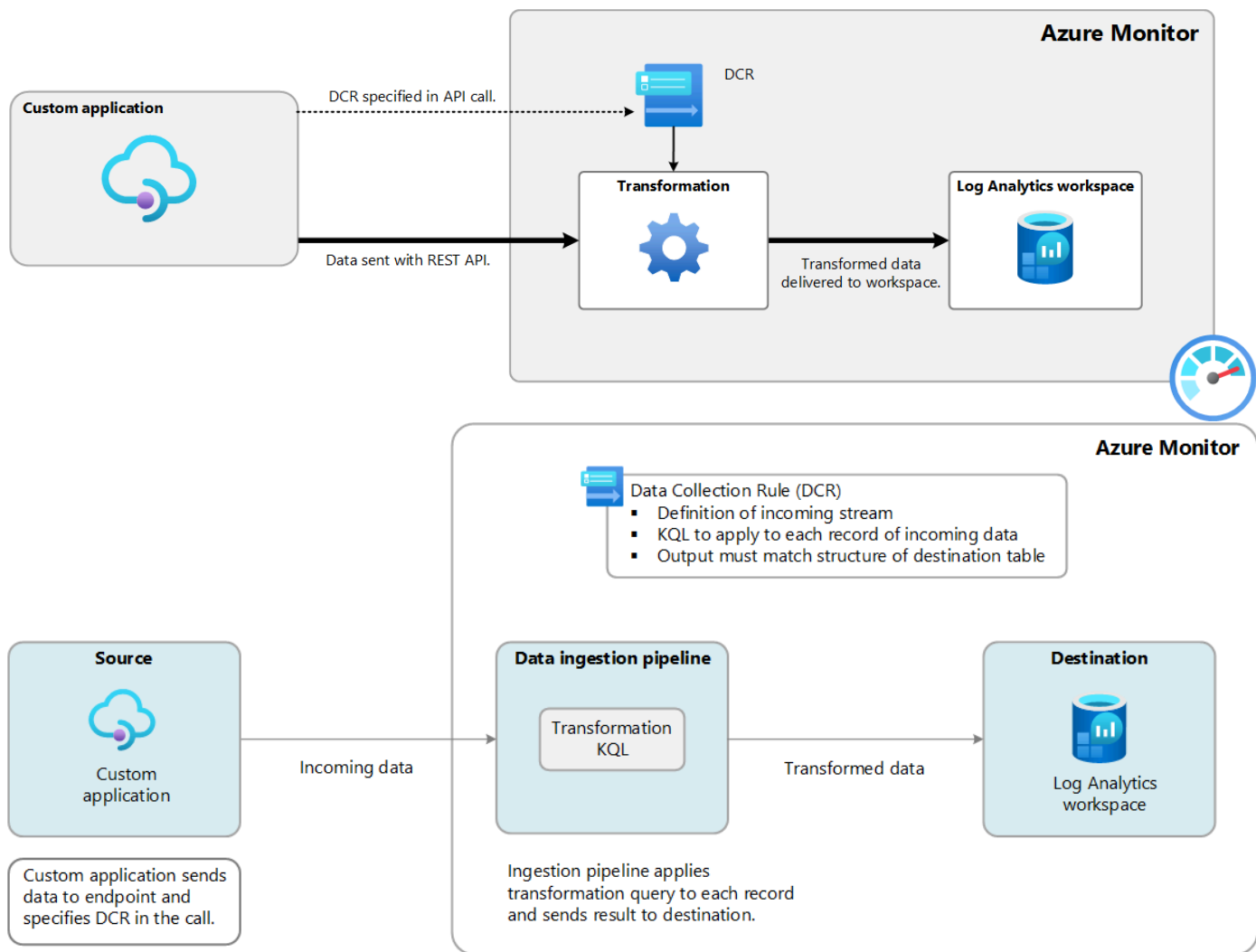
Azure resources automatically collect platform metrics and send them to Azure Monitor Metrics. The following diagram shows the process of using a DCR to send this data to a Log Analytics workspace for analysis using log queries. This process replaces the current method of using [Diagnostic Settings](#) to perform this function.

When the DCR is created, it specifies the workspace and table where the data should be sent. The DCR also includes a transformation that ensures the data is in the correct format for the target table. The DCR is then associated with the resource from which the platform metrics are collected.



Direct ingestion

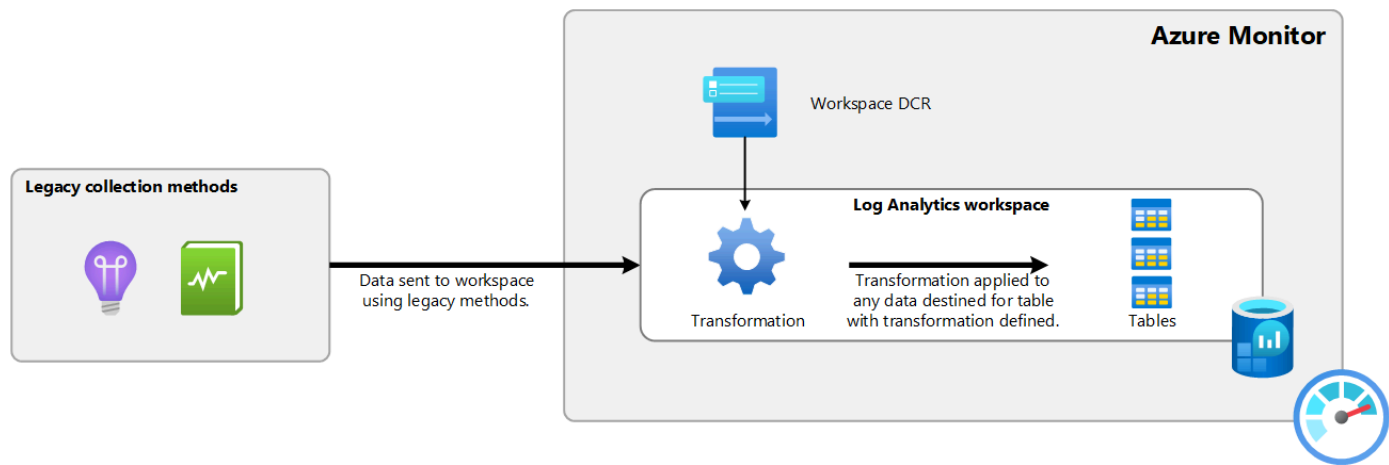
Use direct ingestion to specify a particular DCR to process the incoming data. For example, the following diagram illustrates data from a custom application using Logs ingestion API. Each API call specifies the DCR that processes its data. The DCR understands the structure of the incoming data, includes a transformation that ensures the data is in the format of the target table, and specifies a workspace and table to send the transformed data.



<https://learn.microsoft.com/en-us/azure/azure-monitor/data-collection/data-collection-rule-overview>

Workspace transformation DCR

Workspace transformation DCRs provide transformations for data collection that don't use a DCR. They're applied directly to the Log Analytics workspace. After you create or update the DCR, allow some time for transformations to take effect on incoming data.



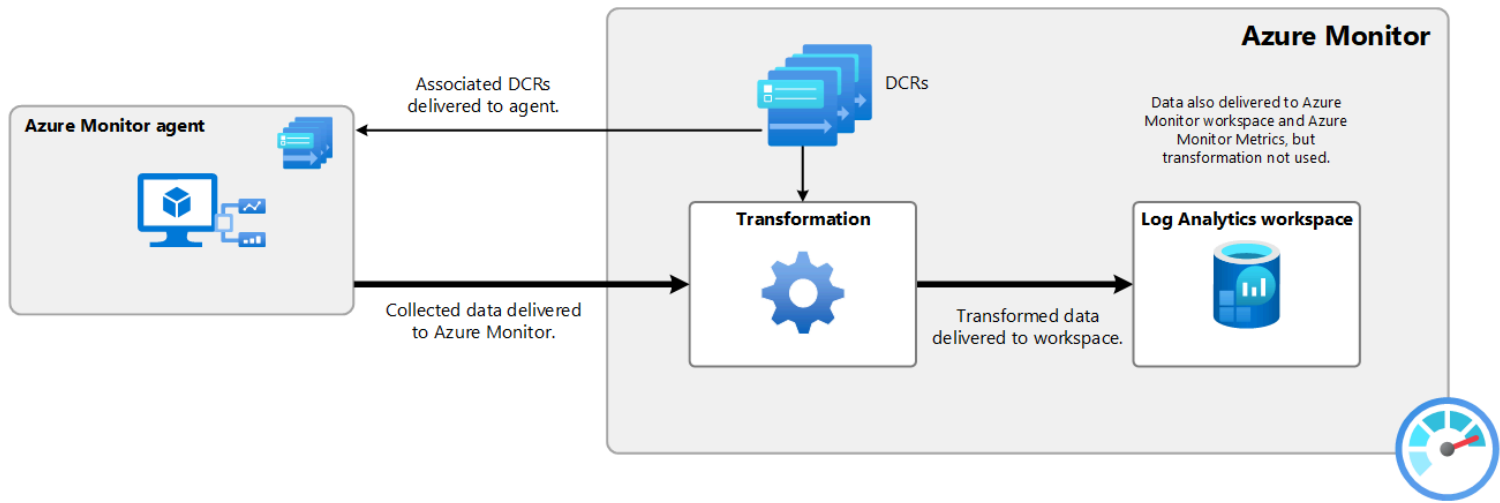
<https://learn.microsoft.com/en-us/azure/azure-monitor/data-collection/data-collection-rule-overview>

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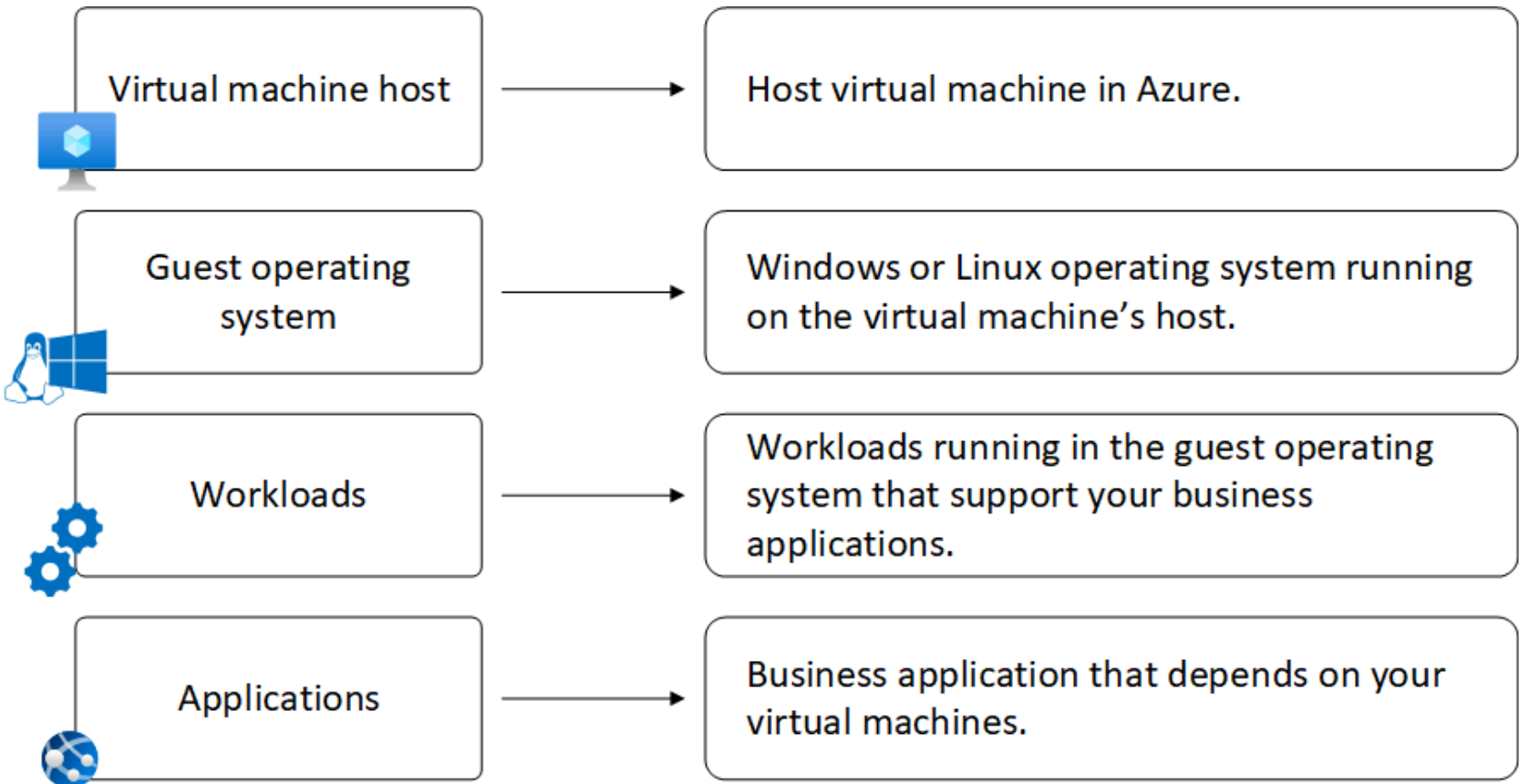


Virtual Machine Monitoring

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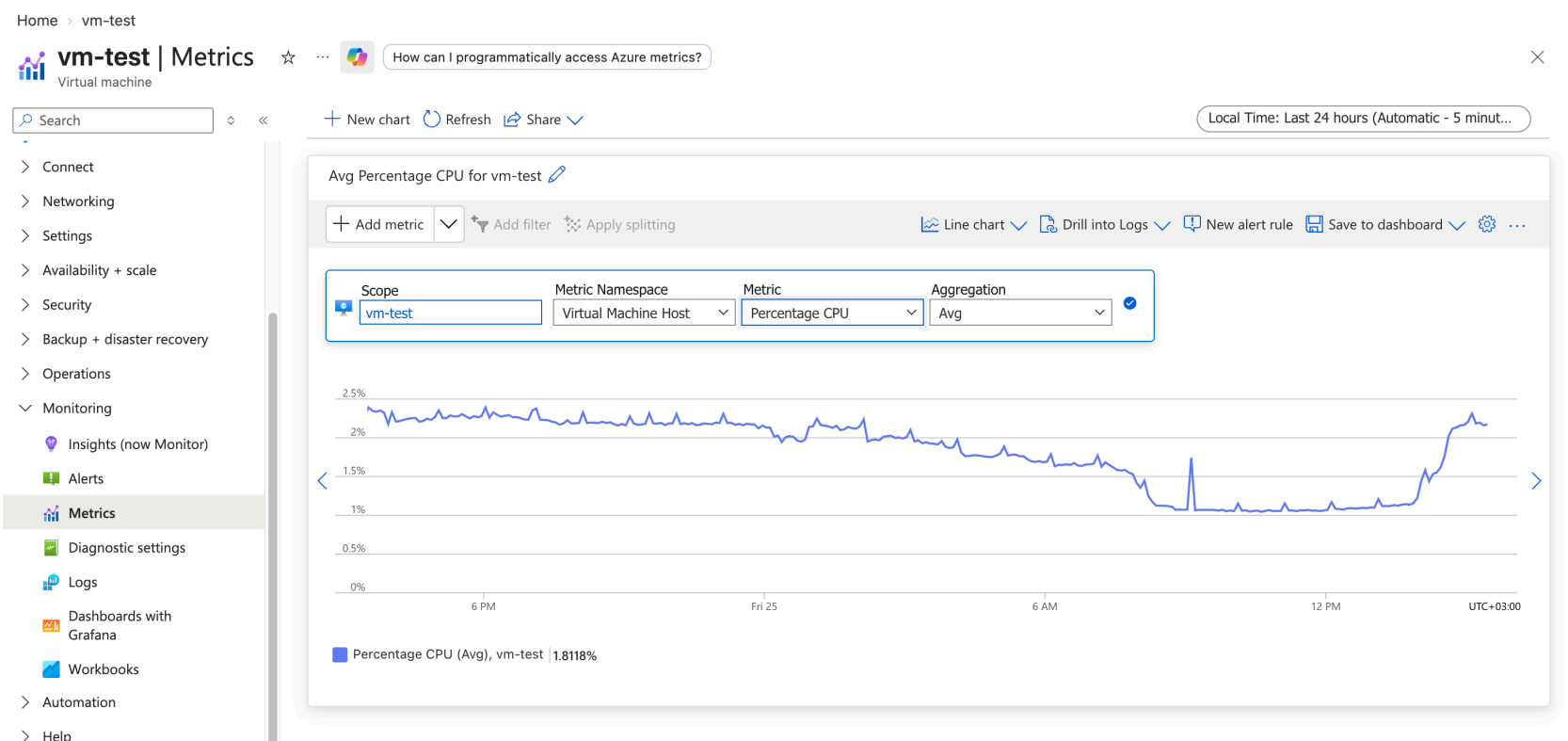
Virtual Machine Monitoring

Azure VMs have several layers that require monitoring. Each of the following layers has a distinct set of telemetry and monitoring requirements.



Azure automatically collects basic metrics for VM hosts. On the VM's Overview page in the Azure portal, you can see built-in graphs for the following important VM host metrics.

Host monitoring:



Guest OS monitoring:

First, I created the LAW, then the DCR. I will determine what data is to be collected and where it will be stored.

Create Log Analytics workspace

Validation passed

Basics

Tags

Review + Create

Log Analytics workspace

by Microsoft

Basics

Subscription

Resource group

Name

Region

Pricing

Pricing tier

The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more](#) about Log Analytics pricing models.

Tags

None

Create Data Collection Rule

Data collection rule management

To access the classic Data Collection Rule creation experience, click here.

Basics

Resources

Collect and deliver

Tags

Review + create

Configure which data sources to collect and where to send the data to.

+ Add new data source

Delete

Data source	Destination type	Destination details
☐ Linux Syslog	Log Analytics Workspaces	law-vm

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

aleyna.erkull@gmail.com

Home > dcr-vm

dcr-vm | DCR visualizer (preview)

Data collection rule

Search

Refresh

Reset Zoom

Feedback

Resource Filters

Subscription == Azure subscription 1

Resource groups : All

Locations : All

Resource Types : All

Show Only Unconnected Resources

Overview

Activity log

Access control (IAM)

Tags

DCR visualizer (preview)

Settings

Configuration

Automation

Security

Monitoring

Help

vm-test

Linux Virtual Machine (VM)

Linux Syslog

Linux Data Source

Linux Syslog

Linux Data Source

law-vm

Log Analytics workspace

When creating the DCR from the portal, I selected the VM I had created myself in the resource section.

The recommended way to install the Azure Monitor Agent through the Azure portal is to create a data collection rule (DCR). The Azure Monitor Agent installs automatically if necessary, and an association is created between the DCR and each machine you select.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

aleyna.erkull@gmail.com

Home > vm-test

vm-test | Extensions + applications

Virtual machine

Search

Extensions

VM Applications

+ Add

Refresh

Update

Enable automatic upgrade

Disable automatic upgrade

Feedback

Search to filter items...

Showing all 1 items

Name	Type	Version	Latest Version	Status	Automatic upgrade st
AzureMonitorLinuxAgent	Microsoft.Azure.Monit...	1.45.0	1.45.0.0	Provisioning succeeded	Disabled

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Disks

Extensions + applications

Operating system

Configuration

law-vm | Logs

Log Analytics workspace

Search

◊ <<

New Query 2 *

New Query 5 * ◊ + ▾

Observability agent

Save ▾

Share ▾ ...

Queries hub

Agent

Ne

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Logs

Resource visualizer

Settings

Syslog ⌄ Time range : Last 3 days Show : 1000 results Add Simple mode ▾

Results Chart

TimeGenerated [UTC] ↑↓

Computer

EventTime [UTC]

Facility

HostName

SeverityLevel

SyslogMessage

> 9/25/2026, 2:47:17.986 AM vm-test 9/25/2026, 2:47:17.0... daemon vm-test info 2026-09-25T02:47:17.986#011Info#011TID:12922945835

> 9/25/2026, 2:47:09.315 AM vm-test Microsoft Azure 47:09.0... daemon vm-test info 2026-09-25T02:47:09.315#011Info#011TID:12922945835

> 9/25/2026, 2:47:03.003 AM vm-test 9/25/2026, 2:47:03.0... daemon vm-test info 2026-09-25T02:47:03.002#011Info#011TID:12922945835

> 9/25/2026, 2:47:00.012 AM vm-test 9/25/2026, 2:47:00.0... daemon vm-test info 2026-09-25T02:47:00.011#011Info#011TID:12922945835

> 9/25/2026, 2:47:00.012 AM vm-test 9/25/2026, 2:47:00.0... daemon vm-test info 2026-09-25T02:47:00.011#011Info#011TID:12922945835

Enable enhanced monitoring for an Azure virtual machine

Virtual machines in Azure automatically send host-level metrics to Azure Monitor, which provide insights into the overall performance and health of the virtual machine. For complete monitoring though, you also need to collect guest-level performance data from the virtual machine, which provides insights into the applications, components, and processes running on the machine and their performance and health.

When you enable enhanced monitoring you choose between two experiences for collecting and visualizing performance data from the guest operating system: OpenTelemetry metrics-based monitoring (recommended for new deployments) and logs-based monitoring (classic).

Unlock enhanced monitoring

Upgrade to OpenTelemetry metrics for more accurate, detailed, and actionable insights into the performance of your VMs and applications. You'll still have access to basic host-level metrics even if you don't upgrade.

Configure

Remind me later

Learn more

Configure monitor | vm-test

Capabilities Review + enable

Infrastructure monitoring

Collect health and performance data from the operating system running on your virtual machine for improved troubleshooting, alerts, and visualizations.

[Customize infrastructure monitoring](#)

Enable detailed metrics

OpenTelemetry metrics [View included metrics](#)

Azure Monitor workspace: (New) defaultazuremonitorworkspace-eus2 (East US 2)

[Classic] Log-based metrics

Data Collection Rule (DCR) (New) msvmi-eastus2-vm-test (East US 2)

Additional Dashboarding

Use built-in Grafana dashboards in the Azure portal to visualize Azure Monitor data with no additional cost or configuration required.

Dashboards with Grafana

At no additional cost

Alerts

Monitor important signals on your virtual machine and get notified quickly about potential issues. Recommended alerts are based on the monitoring configurations selected above and can be customized and managed in the Alerts experience. [Learn more](#)

Enable recommended alerts

Alert me if

Recommended alerts

VM Availability is Less Than 100% (1/1 selected)

OTel Guest OS metric rules (7/7 selected)

vm-test | Monitor

Virtual machine

Search

◊ <<

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Insights (now Monitor)

Alerts

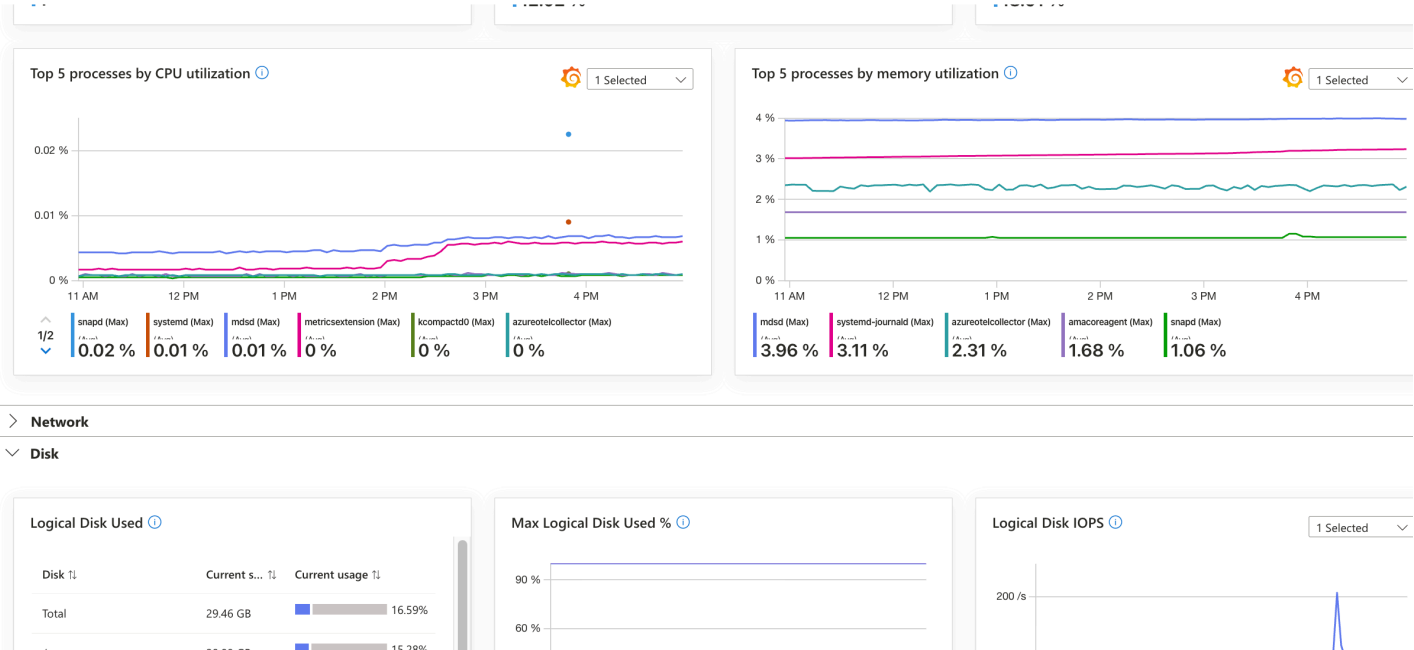
Metrics

Diagnostic settings

Logs

Dashboards with Grafana

Workbooks



Analysis and Response

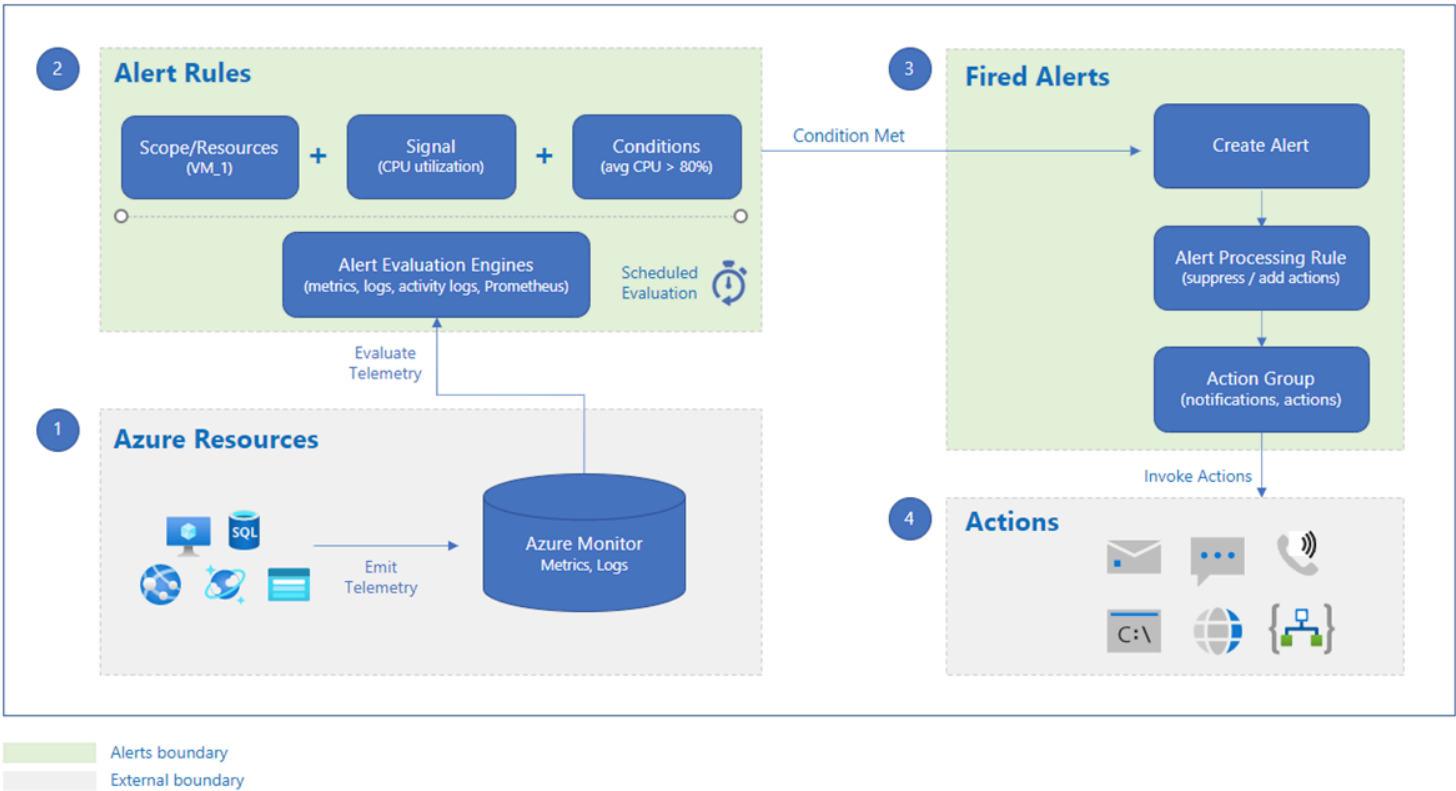


Analysis and Response

Alert Rule

Alerts help you detect and address issues before users notice them by proactively notifying you when Azure Monitor data indicates there might be a problem with your infrastructure or application.

You can alert on any metric or log data source in the Azure Monitor data platform.



Create an alert rule

Scope

Condition

Actions

Details

Tags

Review + create

Product details

Metric alert rule

1 Condition

Terms of use | Privacy statement

Total pricing

0.10 USD/month

Pricing

Scope

Scope level

Subscription

Resource

Azure subscription 1 > rg-terraform-test > vm-test

Condition

Signal name

Percentage CPU

Operator

Greater than

Aggregation type

Average

Threshold value

23

Lookback period

15 minutes

Check every

5 minutes

Home > vm-test | Alerts > Alerts

Alert rules

+ Create

☆ Set up recommended alerts

Columns

Refresh

Export to CSV

Open query

Delete

Enable

Disable

Create an autonomous observability agent: Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. Learn more

Create agent

Search

Subscription : Azure subscription 1

Target resource type : all

Target scope : vm-test

Signal type : all

Severity : all

Status : Enabled

Add tag filter

No grouping

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ cpu-alert-vm	Percentage CPU > 23	2 - Warning	vm-test	Virtual machine	Metrics	Enabled

Kusto Query Language KQL

You use Kusto Query Language (KQL) to write queries in Azure Data Explorer, Azure Monitor Log Analytics, Microsoft Sentinel, and more.

Str:

Table

| operator

| operator

StormEvents
| count

Operation	KQL Operator	Description	
Count rows	count	Returns the number of rows in the table.	
View sample data	take	Returns the specified number of random rows from the table. It can be used to inspect the general structure and content of the data.	

Operation	KQL Operator	Description	
Select specific columns	project	Selects the specific columns to be displayed in the results.	
List unique values	distinct	Lists the unique values in the specified column.	
Filter by condition	where	Filters rows that meet the specified criteria.	
Filter by date/time range	between	Can be used to filter data within a specific value or time range.	
Sort results	sort	Sorts rows according to the specified column. The default sort order is descending (desc); ascending order can be specified using asc .	
Get the first n rows	top	Returns the first n rows after sorting the data by the specified column.	
Create calculated columns	project , extend	project can create a calculated column while selecting the columns to display. extend adds a new calculated column to the existing table columns.	

```
StormEvents
| take 5
| project State, EventType, DamageProperty
```

```
StormEvents
| where State == 'TEXAS' and EventType == 'Flood'
| sort by DamageProperty
| project StartTime, EndTime, State, EventType, DamageProperty
```

Home > vm-test

vm-test | Logs

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

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Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Insights (now Monitor)

Alerts

New Query 2 * ...

Run

Time range : Set in query

Show : 1000 results

```
1 Syslog
2 | where TimeGenerated > ago(1h)
3 | summarize LogCount = count() by bin(TimeGenerated, 10m), SeverityLevel
4 | sort by TimeGenerated desc
```

Results

Chart

TimeGenerated [UTC]	SeverityLevel	LogCount
> 9/25/2026, 3:00:00.000 PM	info	98
> 9/25/2026, 2:50:00.000 PM	info	161
> 9/25/2026, 2:40:00.000 PM	info	125
> 9/25/2026, 2:30:00.000 PM	info	155
> 9/25/2026, 2:20:00.000 PM	info	154
> 9/25/2026, 2:10:00.000 PM	info	133
> 9/25/2026, 2:00:00.000 PM	info	26